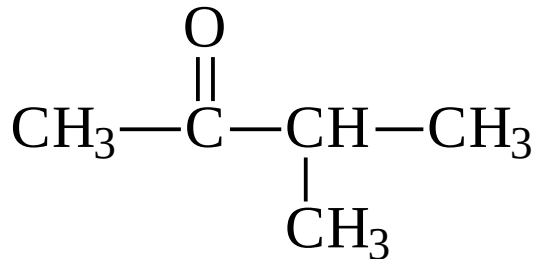
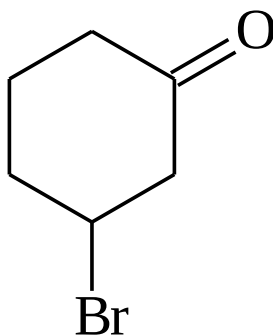


# *Aldehydes and Ketones*

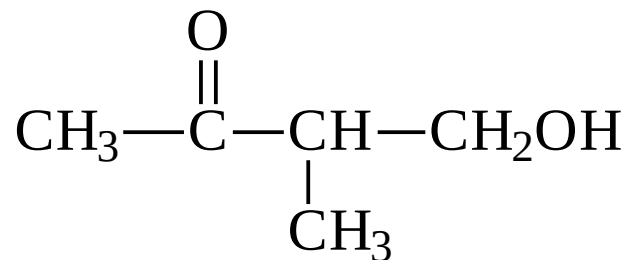
# Examples



3-methyl-2-butanone



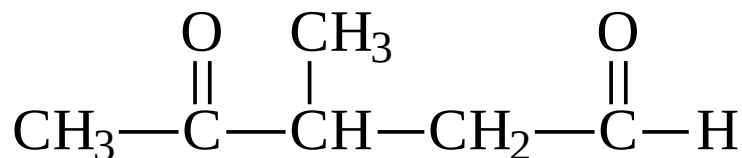
3-bromocyclohexanone



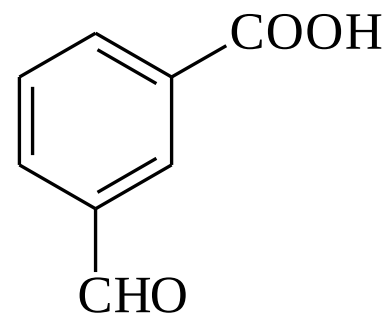
4-hydroxy-3-methyl-2-butanone

## Name as Substituent

- On a molecule with a higher priority functional group, C=O is *oxo-* and -CHO is *formyl*.
- Aldehyde priority is higher than ketone.



3-methyl-4-oxopentanal



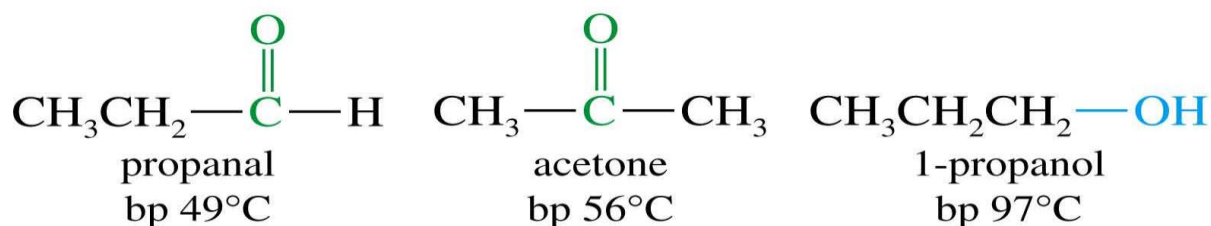
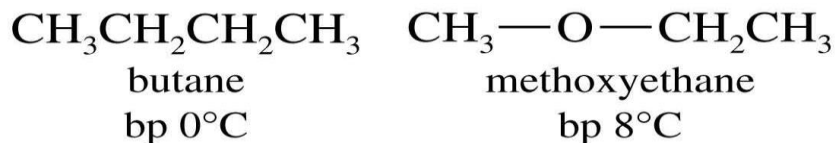
3-formylbenzoic acid 2

# Aldehyde Common Names

- Use the common name of the acid.
- Drop *-ic acid* and add *-aldehyde*.
  - 1 C: formic acid, formaldehyde
  - 2 C's: acetic acid, acetaldehyde
  - 3 C's: propionic acid, propionaldehyde
  - 4 C's: butyric acid, butyraldehyde.

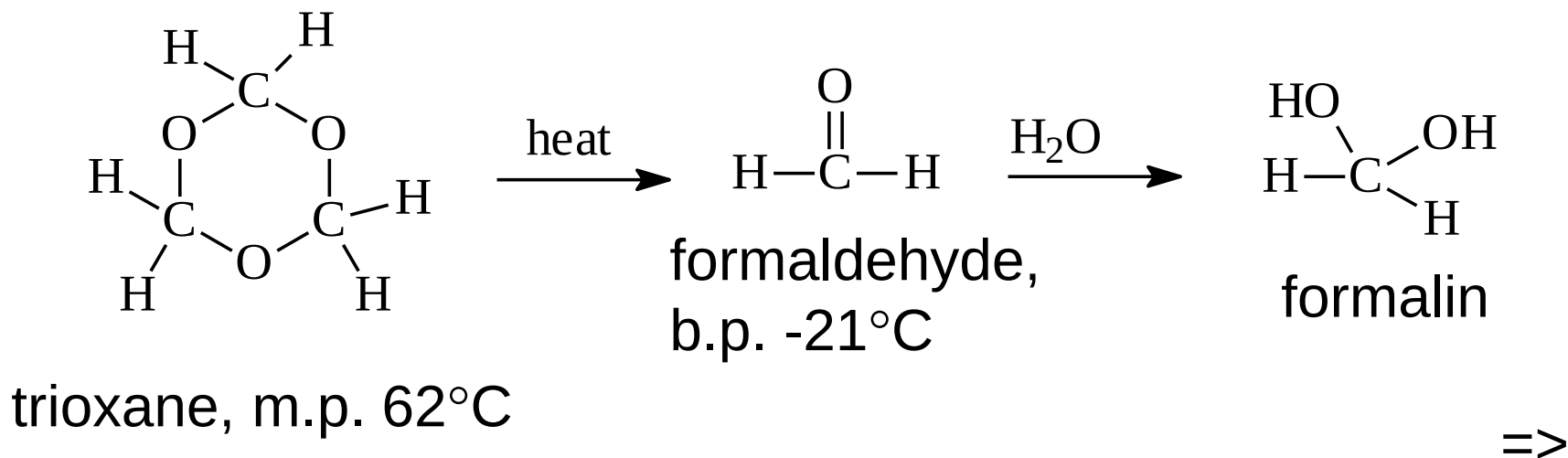
## Boiling Points

- More polar, so higher boiling point than comparable alkane or ether.
- Cannot H-bond to each other, so lower boiling point than comparable alcohol.



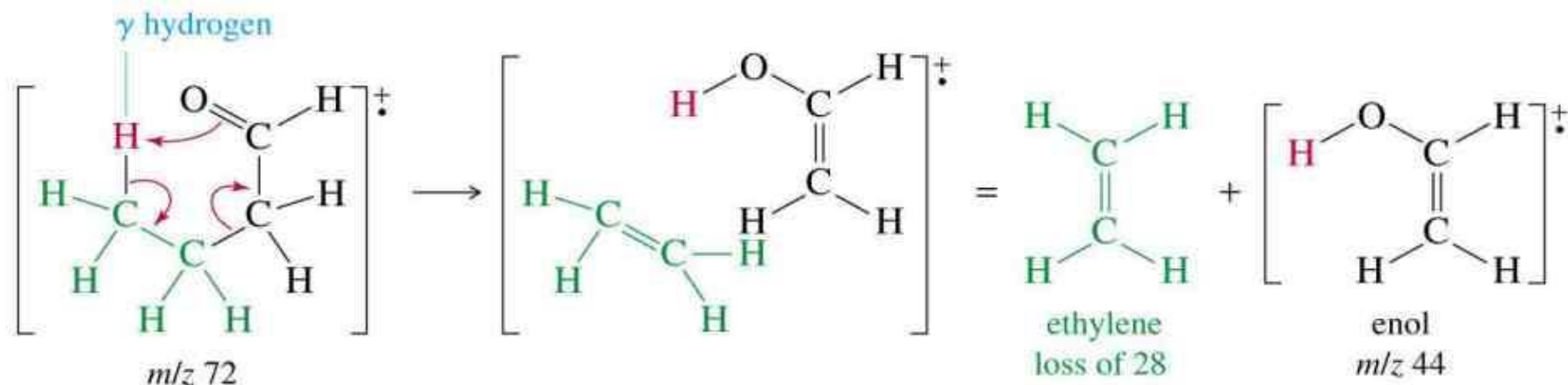
# Formaldehyde

- Gas at room temperature.
- Formalin is a 40% aqueous solution.



# McLafferty Rearrangement

- Loss of alkene (even mass number)
- Must have  $\gamma$ -hydrogen



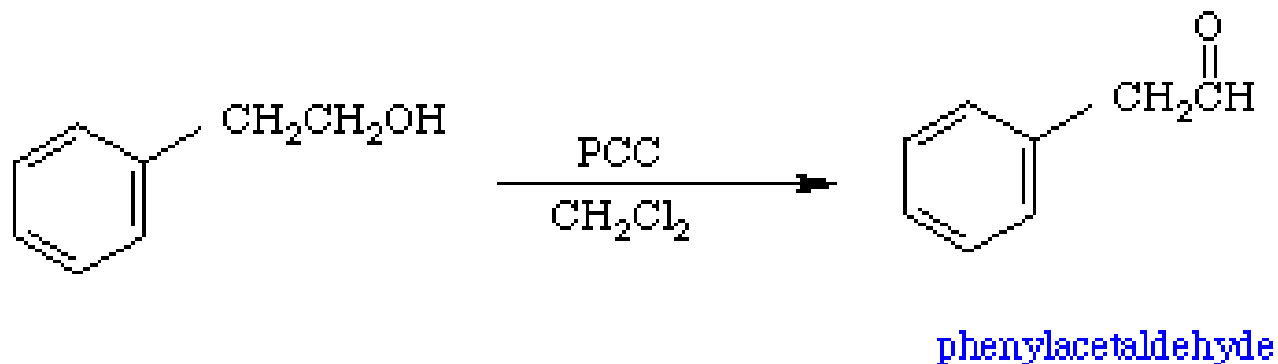
McLafferty rearrangement of butyraldehyde

=>

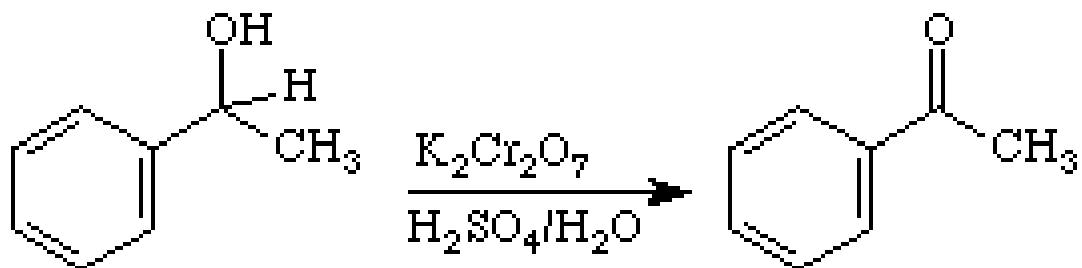
# Synthesis

- Oxidation

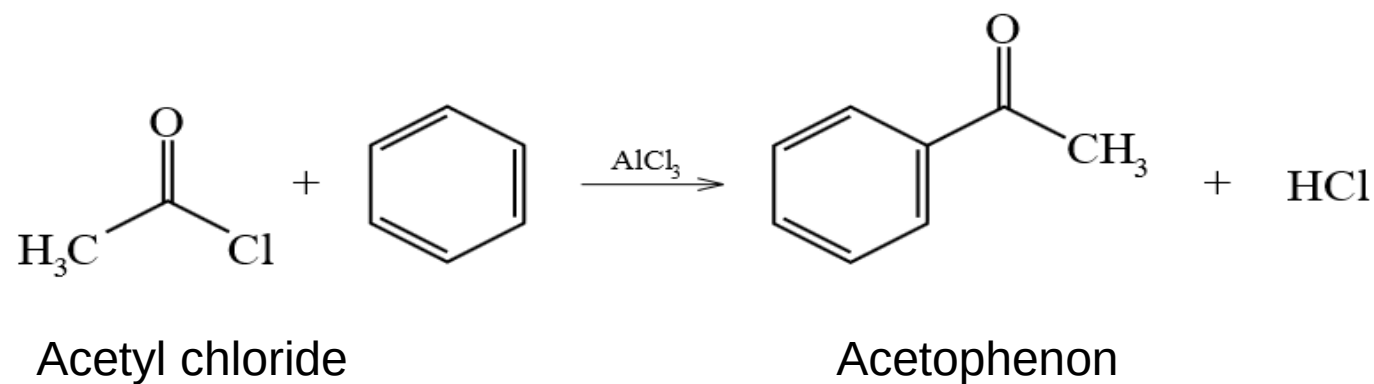
1° alcohol + PCC (Pyridinium chlorochromate) → aldehyde



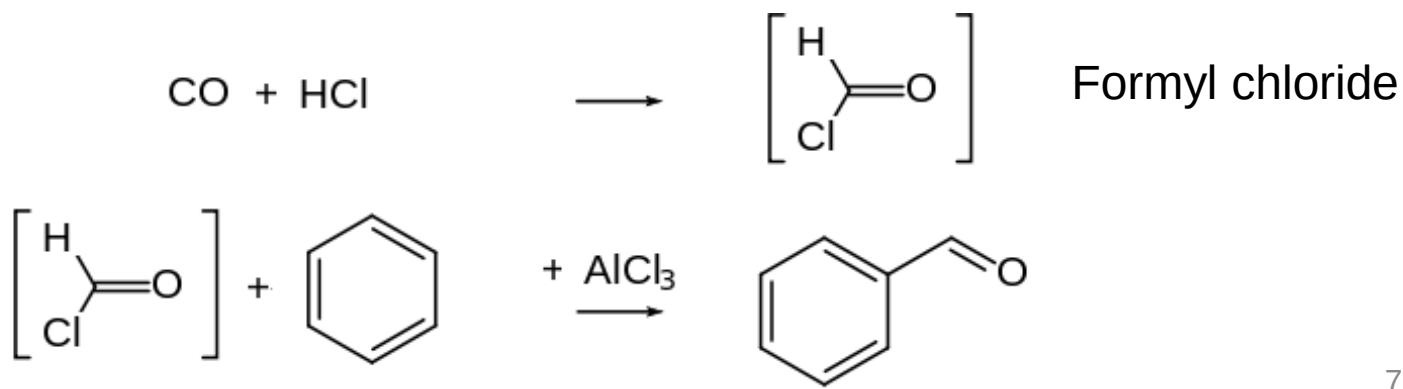
2° alcohol +  $\text{Na}_2\text{Cr}_2\text{O}_7$  /  $\text{K}_2\text{Cr}_2\text{O}_7$  → ketone



- Friedel-Crafts acylation  
 Acid chloride/ $\text{AlCl}_3$  + benzene  $\rightarrow$  ketone



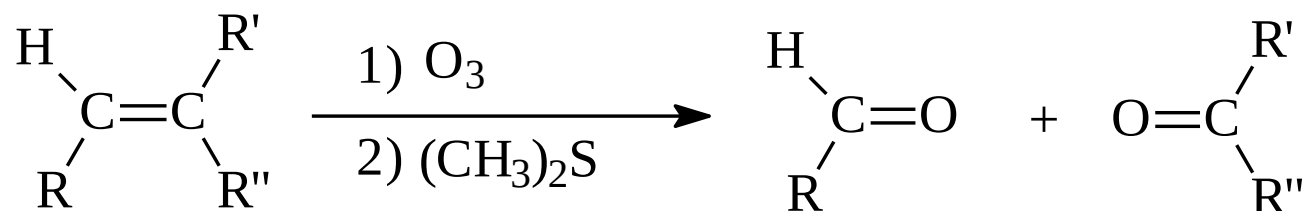
$\text{CO} + \text{HCl} + \text{AlCl}_3/\text{CuCl} + \text{benzene} \rightarrow \text{benzaldehyde}$  (Gatterman-Koch reaction)



## Hydration of terminal alkyne

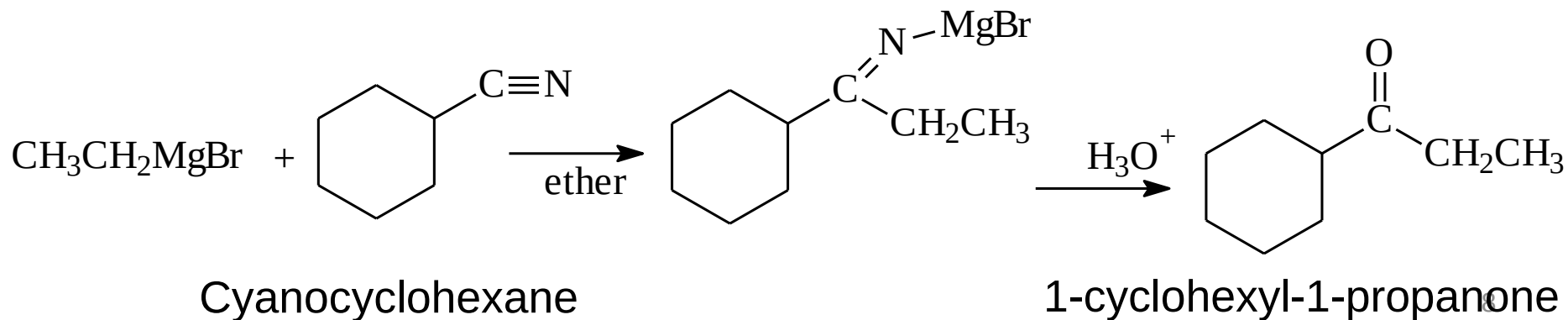
- Use  $\text{HgSO}_4$ ,  $\text{H}_2\text{SO}_4$ ,  $\text{H}_2\text{O}$  for methyl ketone
- Use  $\text{Sia}_2\text{BH}$  (Disiamyl borane) followed by  $\text{H}_2\text{O}_2$  in  $\text{NaOH}$  for aldehyde.

## Ozonolysis of alkenes.



## Ketones from Nitriles

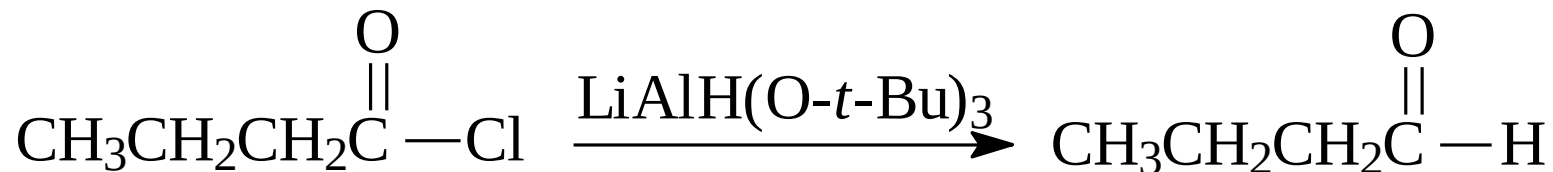
- A Grignard or organolithium reagent attacks the nitrile carbon.
- The imine salt is then hydrolyzed to form a ketone.





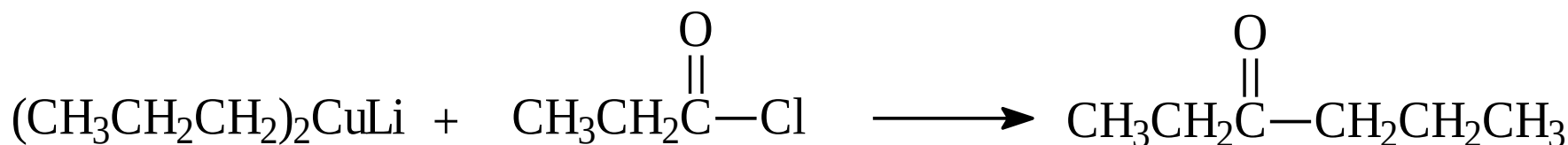
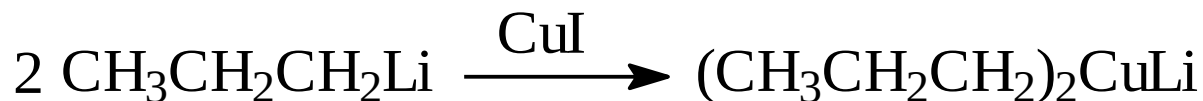
## Aldehydes from Acid Chlorides

Use a mild reducing agent to prevent reduction to primary alcohol.



## Ketones from Acid Chlorides

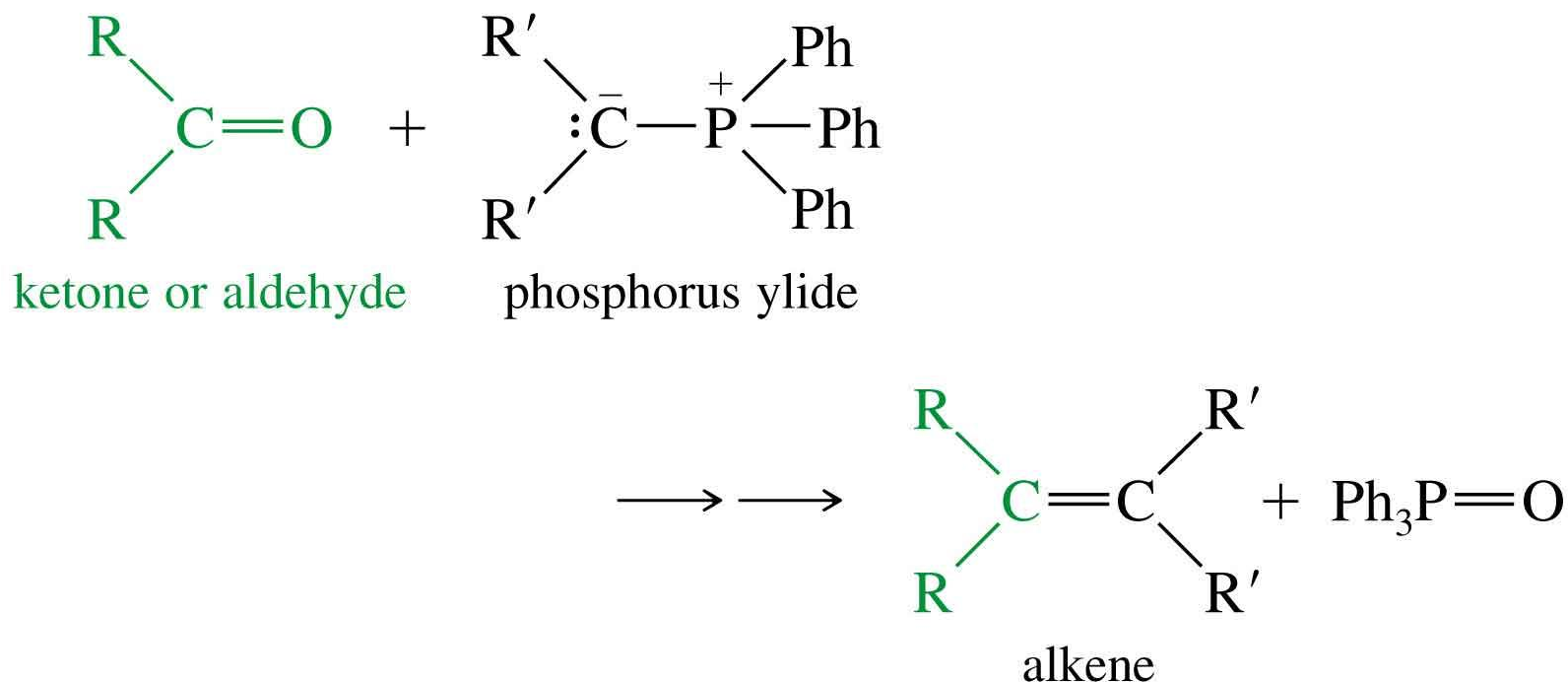
Use lithium dialkylcuprate ( $\text{R}_2\text{CuLi}$ ), formed by the reaction of 2 moles of  $\text{R-Li}$  with cuprous iodide.



# Wittig Reaction

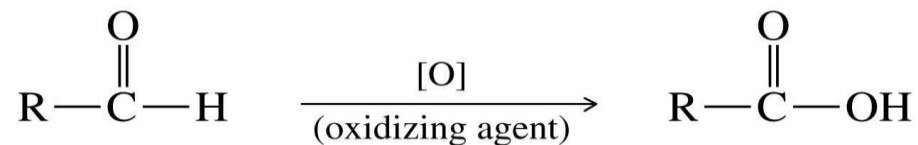
- Nucleophilic addition of phosphorus ylides.
- Product is alkene. C=O becomes C=C.

*The Wittig reaction*

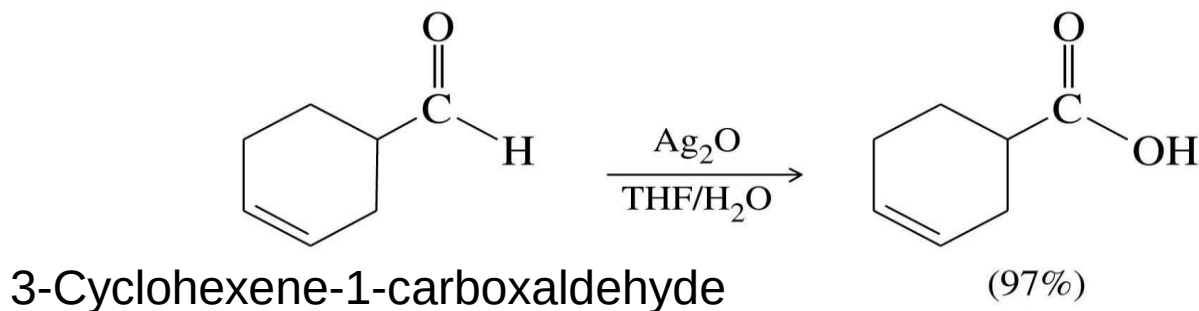
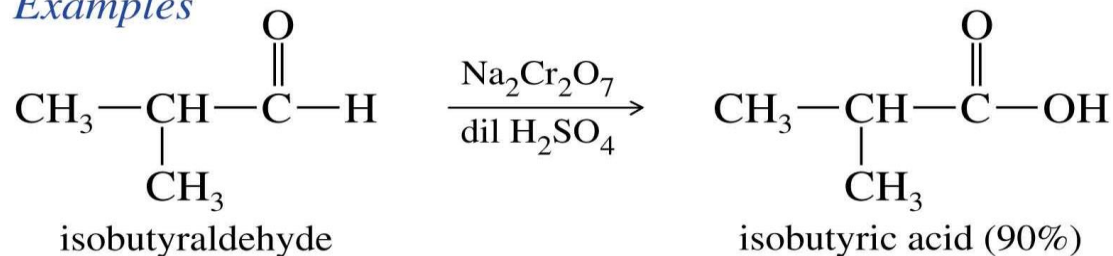


# Oxidation of Aldehydes

Easily oxidized to carboxylic acids.

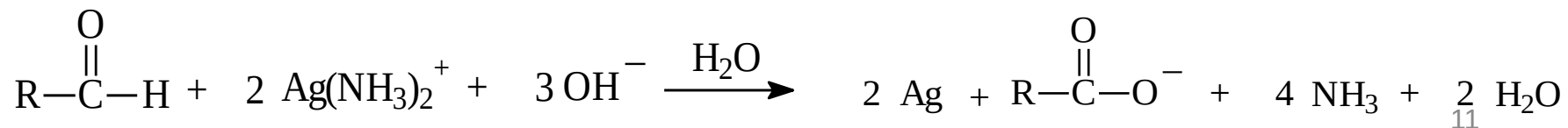


*Examples*



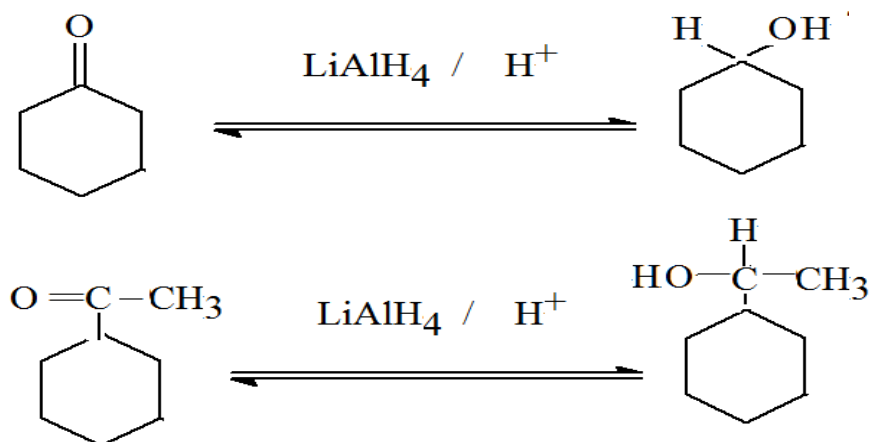
## Tollens Test

- Aldehyde reaction forms a silver mirror.



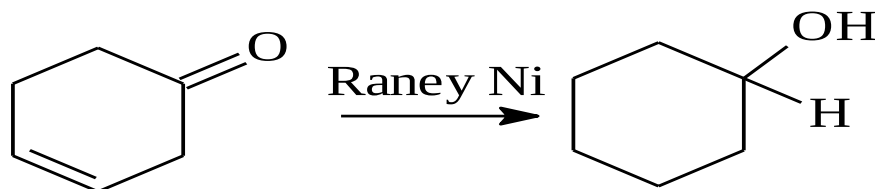
# Reduction Reactions of Aldehydes & Ketones

- Sodium borohydride,  $\text{NaBH}_4$ , reduces  $\text{C}=\text{O}$ , but not  $\text{C}=\text{C}$ .
- Lithium aluminum hydride,  $\text{LiAlH}_4$ , much stronger, difficult to handle.
- Hydrogen gas with catalyst also reduces the  $\text{C}=\text{C}$  bond.



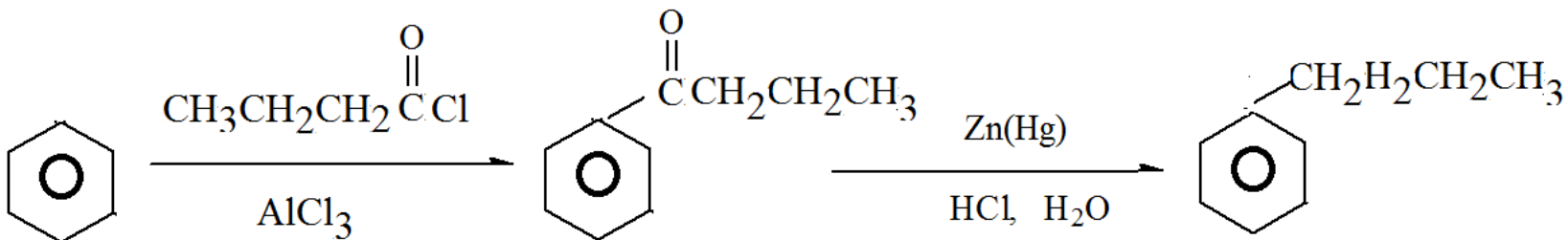
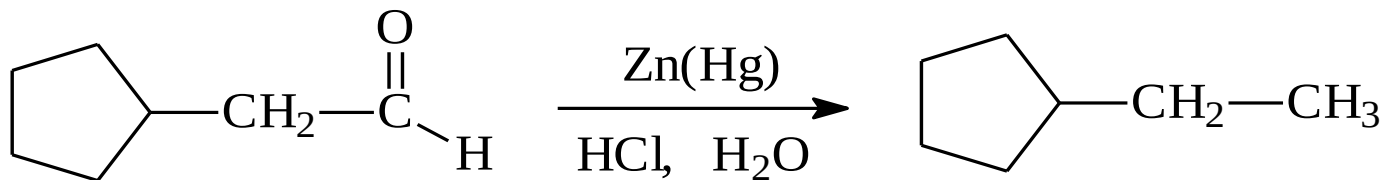
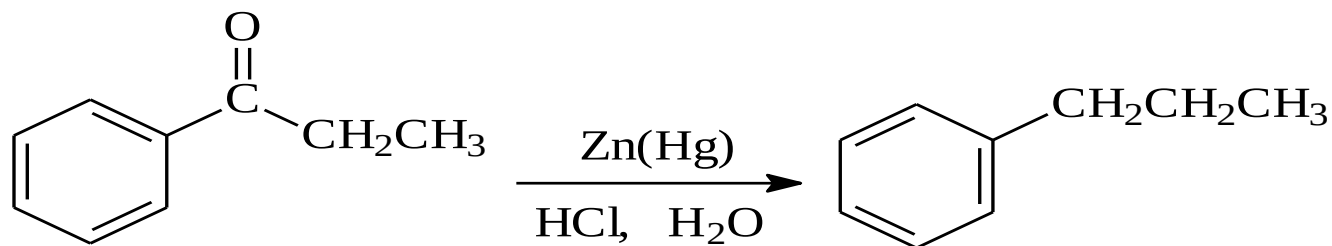
## Catalytic Hydrogenation

- Widely used in industry.
- Raney nickel, finely divided Ni powder saturated with hydrogen gas.
- Pt and Rh also used as catalysts.



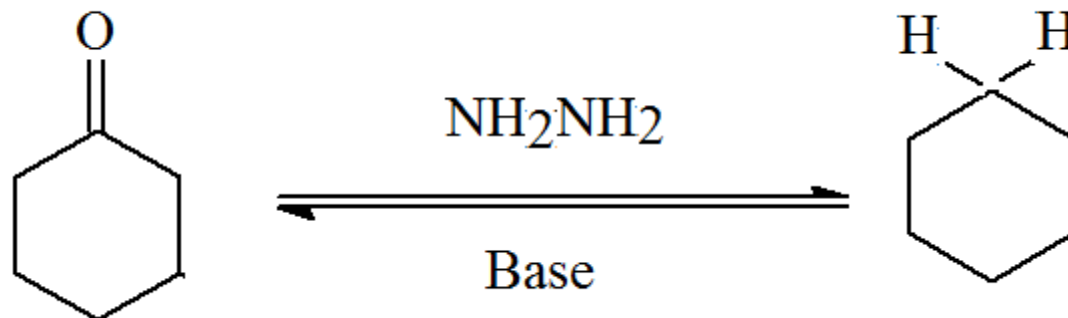
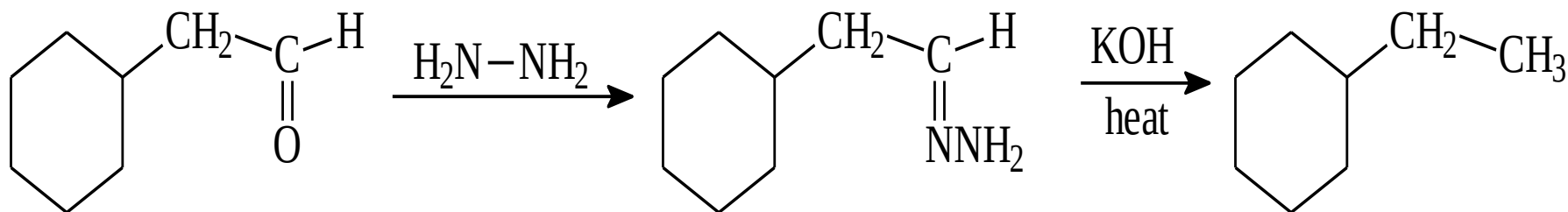
# Reduction of Hydrocarbon -Deoxygenation

- Reduction of C=O to CH<sub>2</sub>.
- Two methods:
  - Clemmensen reduction if molecule is stable in hot acid.



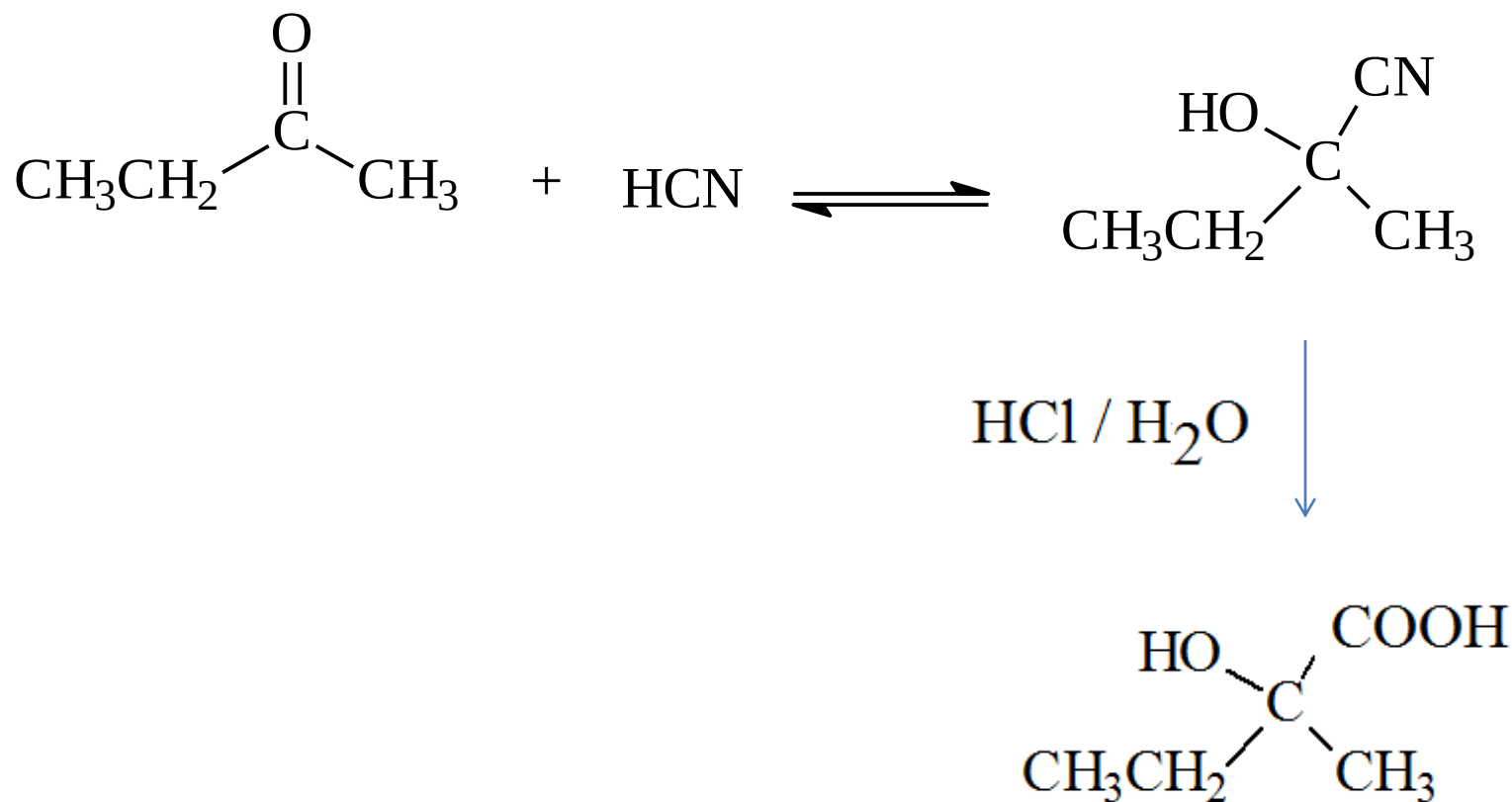
# Wolff-Kishner reduction

if molecule is stable in very strong base. Form hydrazone, then heat with strong base like KOH or potassium *t*-butoxide.

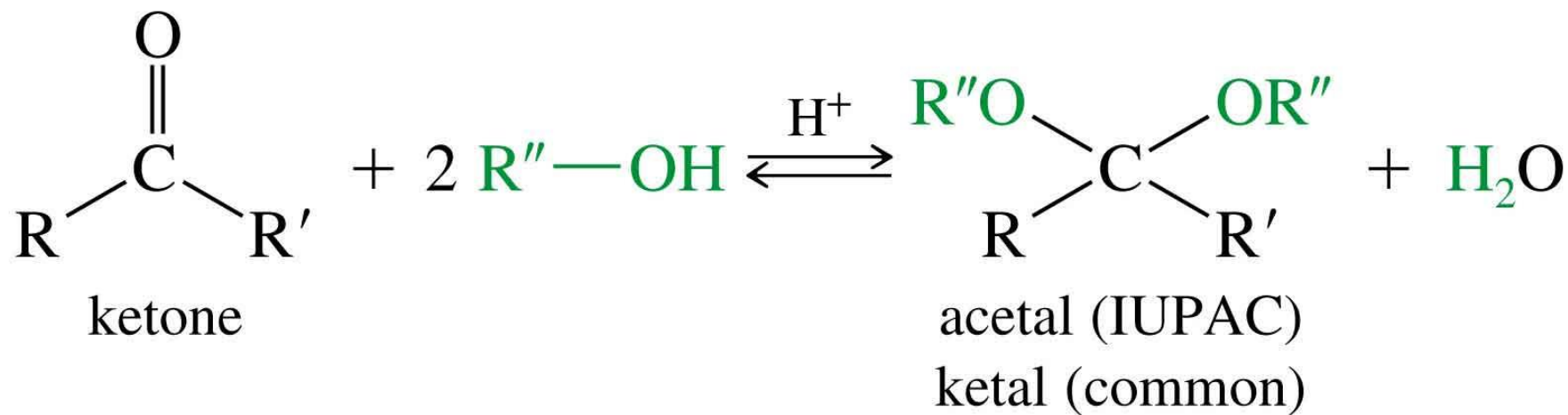
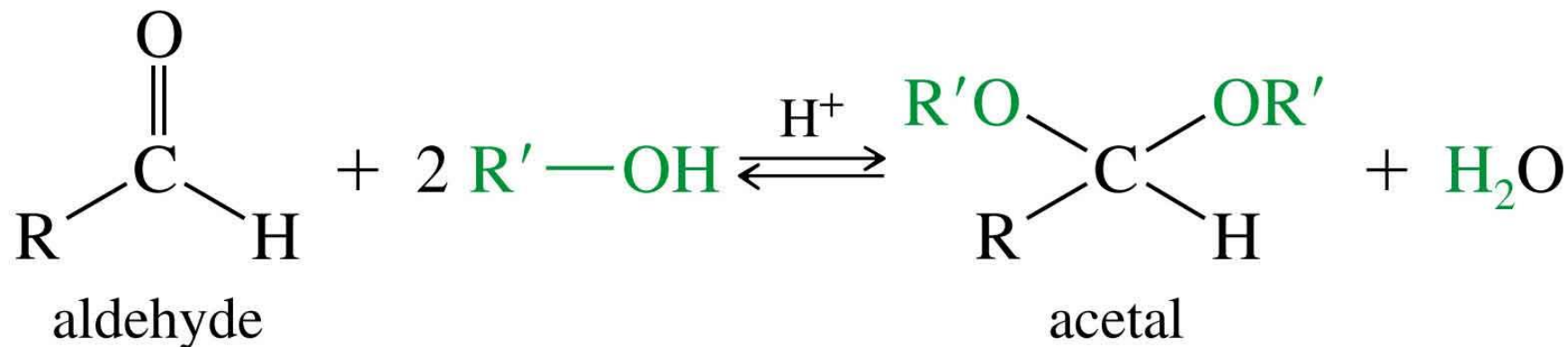


# Addition of Cyanide; Cyano – hydrin formation

- Use NaCN, HCN or KCN in base to add cyanide, then protonate to add H.
- Reactivity formaldehyde > aldehydes > ketones >> bulky ketones.



# Addition of Alcohols

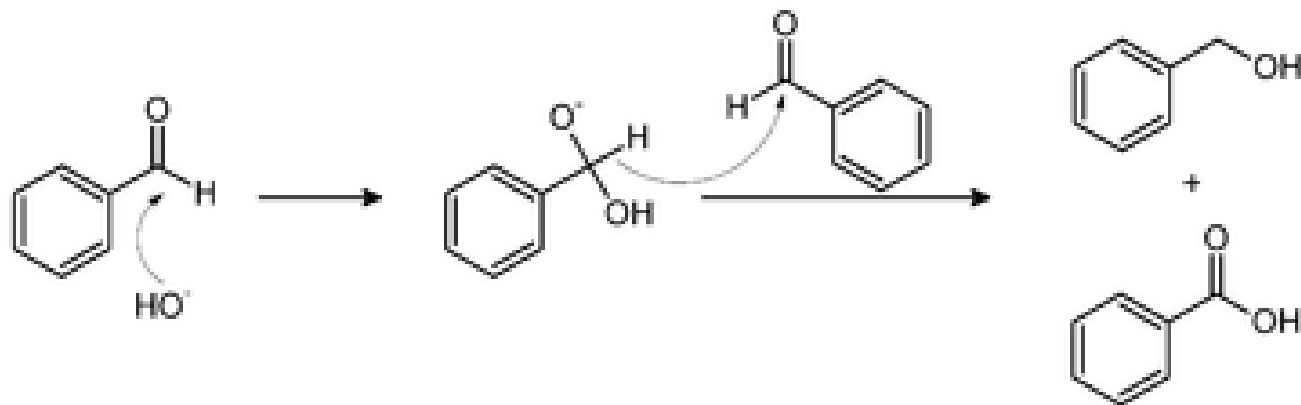


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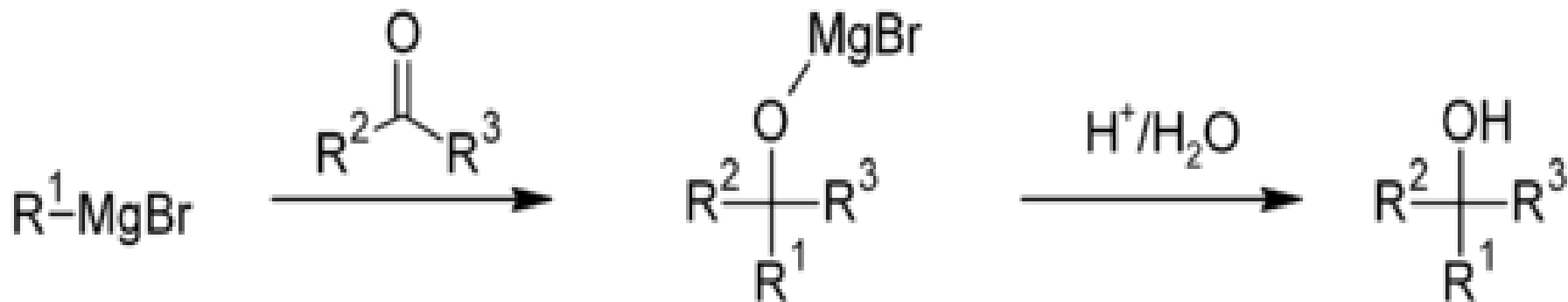


# Cannizzaro reaction

2 molecules of aldehydes converts to a carboxylic acid (salt) and an alcohol

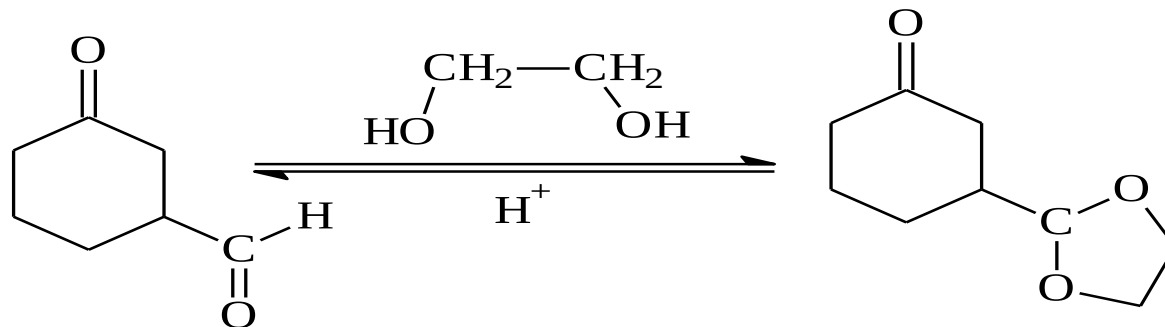


# Grignard reaction



# Selective Reaction of Ketone

- Hydrolyze easily in acid, stable in base.
- Aldehydes more reactive than ketones.



- React with strong nucleophile (base)
- Remove protective group.

